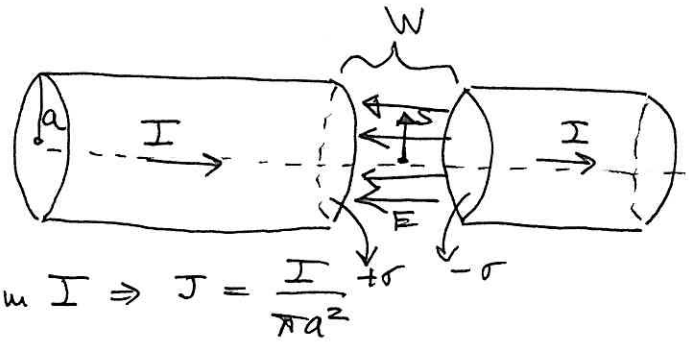
$\boxed{7.31}$



Uniform $I \Rightarrow J = \dfrac{I}{\pi a^2}$

Clearly only AC (time-dependent) current can flow
$\Rightarrow \vec{B}$ is induced by displacement current $\vec{J}_d = \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t}$

Width $w \ll a \Rightarrow$ plate capacitor with
uniform $\vec{E} = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 \cdot \pi a^2}$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

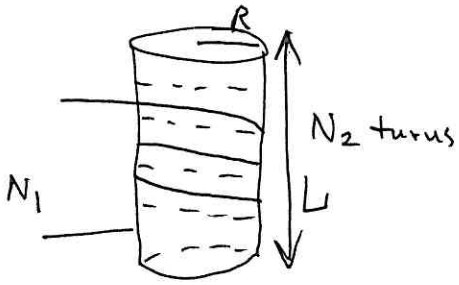
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \underbrace{I_{enc}}_{\substack{\| \\ 0 \\ \text{inside the gap}}} + \mu_0 \varepsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{A}$$

Amperian loop
of radius s $\quad B \cdot 2\pi s = \mu_0 \varepsilon_0 \dfrac{\partial E}{\partial t} \cdot \pi s^2 \quad \leftarrow$ flux through area πs^2

$$\Rightarrow B = \frac{\mu_0 \varepsilon_0}{2\pi s} \pi s^2 \frac{\partial}{\partial t}\left(\frac{Q}{\varepsilon_0 \pi a^2} \right) = \frac{\mu_0 \varepsilon_0}{2} s \frac{I}{\varepsilon_0 \pi a^2} = \frac{\mu_0 I s}{2\pi a^2}$$

but $\dfrac{\partial Q}{\partial t} = I =$ current!

7.53



In this geometry all flux from primary coil is passing through secondary.

Current I in primary, $N_1 = n_1 L$ = total number of turns

Produces $B = \mu_0 n_1 I$

Hence flux $\phi_2 = \pi R^2 (\mu_0 n_1 I) \cdot N_2$ through the secondary coil.

Flux through the primary coil $\phi_1 = \pi R^2 (\mu_0 n_1 I) \cdot N_1$

Back emf in the primary $\mathcal{E}_1 = -\frac{\partial \phi_1}{\partial t} = -\mu_0 \pi R^2 \frac{N_1^2}{L} \frac{dI}{dt}$

Induced emf in the secondary $\mathcal{E}_2 = -\frac{\partial \phi_2}{\partial t} = -\mu_0 \pi R^2 \frac{N_1 N_2}{L} \frac{dI}{dt}$

$\Rightarrow$ the ratio $\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_1 N_2 / L}{N_1^2 / L} = \frac{N_2}{N_1}$.

7.54

From P7.53 $L_1 = \mu_0 \pi R^2 \frac{N_1^2}{L}$ (here N_1 = total number of turns, L = total length)

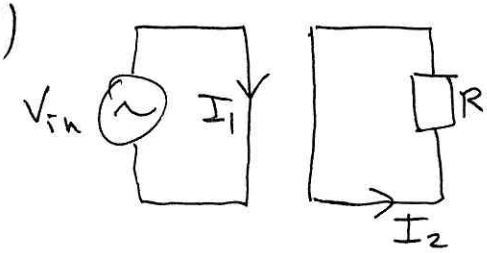
Hence $L_2 = \mu_0 \pi R^2 \frac{N_2^2}{L}$

We found (P7.53) $M = \mu_0 \pi R^2 \frac{N_1 N_2}{L}$

(a) Thus $M = \mu_0 \pi R^2 \frac{N_1 N_2}{L} = \mu_0 \pi R^2 \sqrt{\frac{N_1^2}{L} \frac{N_2^2}{L}} =$

$= \sqrt{L_1 L_2}$.

(b)



Total emf in primary circuit (1):

$$\mathcal{E}_1 = V_{in} - L_1 \frac{dI_1}{dt} - M \frac{dI_2}{dt}$$

In secondary $\mathcal{E}_2 = -L_2 \frac{dI_2}{dt} - M \frac{dI_1}{dt}$

$$\Rightarrow \quad L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt} = V_1 \cos(\omega t) \qquad \text{(no resistors in 1)}$$

$$L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt} = -I_2 R$$

(c) Solve generalized system with complex $e^{i\omega t}$ time variation and then take real parts to obtain physical currents and voltages:

$$\begin{cases} L_1 \dfrac{dI_1}{dt} + M \dfrac{dI_2}{dt} = V_1 e^{i\omega t} \\ L_2 \dfrac{dI_2}{dt} + M \dfrac{dI_1}{dt} = -I_2 R \end{cases}$$

let $I_1(t) = I_1 e^{i\omega t}$; $I_2(t) = I_{20} + I_2 e^{i\omega t}$

$$\frac{dI_1}{dt} = i\omega I_1 e^{i\omega t}$$

$$\begin{cases} L_1 (i\omega) I_1 e^{i\omega t} + M (i\omega) I_2 e^{i\omega t} = V_1 e^{i\omega t} \\ L_2 (i\omega) I_2 e^{i\omega t} + M (i\omega) I_1 e^{i\omega t} = -\left(I_{20} + I_2 e^{i\omega t}\right) R \end{cases}$$

Thus it must be that $I_{20} = 0$.

Divide by $e^{i\omega t}$:

$$\begin{cases} L_1 I_1 + M I_2 = \dfrac{V_1}{i\omega} \Rightarrow I_2 = \left(\dfrac{V_1}{i\omega} - L_1 I_1\right) \dfrac{1}{M} \\ L_2 I_2 + M I_1 = -\dfrac{I_2 R}{i\omega} \end{cases}$$

$$\frac{L_2}{M}\left(\frac{V_1}{i\omega} - L_1 I_1\right) + M I_1 = -\frac{R}{i\omega}\left(\frac{V_1}{i\omega} - L_1 I_1\right)\frac{1}{M}$$

$$\underbrace{M^2 I_1 - L_2 L_1 I_1}_{0} - \frac{R}{i\omega} L_1 I_1 = -\left(\frac{L_2}{\cancel{M}} + \frac{R}{i\omega}\right)\frac{V_1}{i\omega}$$

$$\Rightarrow L_1 I_1 = \left(\frac{L_2}{\cancel{M}} + \frac{R}{i\omega}\right)\frac{V_1}{R}$$

$M^2 = L_1 L_2$

$$I_1 = \frac{V_1}{R}\left(\frac{L_2}{L_1} - \frac{iR}{\omega L_1}\right)$$

$$I_2 = \frac{1}{M}\frac{V_1}{i\omega} - \frac{1}{M}\left(\frac{L_2}{} + \frac{R}{i\omega}\right)\frac{V_1}{R} = -\frac{L_2}{M}\frac{V_1}{R}$$

$$\Rightarrow I_1(t) = \mathrm{Re}\left\{\frac{V_1}{R}\left(\frac{L_2}{L_1} - \frac{iR}{\omega L_1}\right)e^{i\omega t}\right\} = \left(e^{i\omega t} = \cos\omega t + i\sin\omega t\right)$$

$$= \frac{V_1}{R}\left(\frac{L_2}{L_1}\cos\omega t + \frac{R}{\omega L_1}\sin\omega t\right) =$$

$$= \frac{V_1}{L_1}\left(\frac{1}{\omega}\sin\omega t + \frac{L_2}{R}\cos\omega t\right).$$

$$I_2(t) = \mathrm{Re}\left(-\frac{L_2}{M}\frac{V_1}{R}e^{i\omega t}\right) = -\frac{L_2 V_1}{MR}\cos\omega t.$$

(d) $$\frac{V_{out}}{V_{in}} = \frac{I_2 R}{V_1\cos\omega t} = \frac{-\frac{L_2 V_1}{M}\cos\omega t}{V_1\cos\omega t} = -\frac{L_2}{M} = -\frac{N_2}{N_1}.$$

The $(-)$ sign just shows that the two voltages are out of phase.

(c) Input power $P_{in} = V_{in} I_1 = \frac{V_1}{L_1}\left(\frac{1}{\omega}\sin\omega t + \frac{L_2}{R}\cos\omega t\right)\cdot V_1\cos\omega t$

Average over the period $T = \frac{2\pi}{\omega}$:

$$\langle P_{in}\rangle = \frac{1}{T}\int_0^{2\pi/\omega} dt\ \frac{V_1^2}{L_1}\left(\frac{1}{\omega}\sin\omega t\cos\omega t + \frac{L_2}{R}\cos^2\omega t\right) =$$

$$= (x = \omega t) = \frac{1}{2\pi}\int_0^{2\pi} dx\ \frac{V_1^2}{L_1}\left(\frac{\sin x\cos x}{\omega} + \frac{L_2}{R}\cos^2 x\right) =$$

$$= \frac{1}{2\pi}\frac{V_1^2}{L_1}\frac{L_2}{R}\cdot\pi = \frac{1}{2}\frac{V_1^2 L_2}{R\ L_1}\quad\left(\text{because } \int_0^{2\pi} dx\ \sin x\cos x = 0\right).$$

$$\langle P_{out} \rangle = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} dt \left(-\frac{L_2 V_1}{MR} \cos\omega t \right) \left(-\frac{N_2}{N_1} \cdot V_1 \cos\omega t \right)$$

$$= \frac{1}{2\pi} \frac{V_1^2}{R} \frac{L_2^2}{M} \underbrace{\int_0^{2\pi} dx \cos^2 x}_{\pi} = \frac{1}{2} \frac{V_1^2}{R} \frac{L_2^2}{L_1 L_2} =$$

$$= \frac{1}{2} \frac{V_1^2}{R} \frac{L_2}{L_1}.$$

Thus $\langle P_{in} \rangle = \langle P_{out} \rangle \Rightarrow$ energy is conserved!

$\boxed{9.2}$ Standing wave $f(z,t) = A \sin(kz) \cos(kvt) =$

$$= \frac{1}{2} A \Big\{ \underbrace{\sin(kz + kvt)}_{\text{left-moving wave}} + \underbrace{\sin(kz - kvt)}_{\text{right-moving wave}} \Big\}$$

Left-moving and right-moving waves satisfy wave equation $\Rightarrow$ the standing wave does so as well.

Explicitly: $$\frac{\partial^2 f}{\partial z^2} = -A k^2 \sin(kz) \cos(kvt)$$

$$\frac{\partial^2 f}{\partial t^2} = -A k^2 v^2 \sin(kz) \cos(kvt)$$

$$\Rightarrow \quad \frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}.$$

$\boxed{9.3}$ Express A_3, δ_3 from: $A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}$.

$$(A_3 e^{i\delta_3})(A_3 e^{-i\delta_3}) = |A_3|^2 = (A_1 e^{i\delta_1} + A_2 e^{i\delta_2})(A_1 e^{-i\delta_1} + A_2 e^{-i\delta_2})$$

$$= A_1^2 + A_2^2 + A_1 A_2 \underbrace{\left(e^{i(\delta_1 - \delta_2)} + e^{-i(\delta_1 - \delta_2)} \right)}_{2\cos(\delta_1 - \delta_2)} = A_1^2 + A_2^2 + 2 A_1 A_2 \cos(\delta_1 - \delta_2)$$

$$\Rightarrow A_3 = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\delta_1 - \delta_2)}$$

Then $e^{i\delta_3} = \frac{A_1}{A_3} e^{i\delta_1} + \frac{A_2}{A_3} e^{i\delta_2}$ $\Rightarrow$ separate real and imaginary parts on both sides:

$$\begin{cases} \cos\delta_3 = \frac{A_1}{A_3}\cos\delta_1 + \frac{A_2}{A_3}\cos\delta_2 \\ \sin\delta_3 = \frac{A_1}{A_3}\sin\delta_1 + \frac{A_2}{A_3}\sin\delta_2 \end{cases}$$

$$\Rightarrow \tan(\delta_3) = \frac{A_1\sin\delta_1 + A_2\sin\delta_2}{A_1\cos\delta_1 + A_2\cos\delta_2}.$$